Ojas Srivastava

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Software Engineer Intern — Monomer Research

B.Tech Artificial Intelligence, SVNIT Surat | **Batch 2028** (Pre-final) | CGPA **9.20/10**

Modern C++ | Linux | DSA | OS | Computer Architecture | Concurrency | Networking

Summary

Pre-final-year **B.Tech** student (Batch **2028**), CGPA **9.20/10**, pursuing **Software Engineer Intern** at Monomer Research. Strong **C++** programmer (STL, C++17) who builds and optimizes **high-performance, reliable** software on **Linux**. Solid foundation in **Data Structures & Algorithms**, **Operating Systems**, and **Computer Organization & Architecture**; hands-on work on **latency**-sensitive systems, **concurrency**, and real-time **networking**. Experienced **debugging**, **testing**, and **optimizing** production code through profiling, automated validation, and peer **code review**. Strong analytical and problem-solving skills; excited to contribute to resilient, low-latency platform engineering.

Education

Bachelor of Technology in Artificial Intelligence (CS-related) — CGPA: 9.20/10

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat, India — *Aug 2024 – May 2028*

- Pre-final year (3rd year), graduating **May 2028**; CGPA **9.20/10**; Computer Science–related discipline (Artificial Intelligence)
- **Coursework: Data Structures & Algorithms**, **Operating Systems**, **Computer Organization & Architecture**, OOP (C++), Computer Networks (TCP/IP), DBMS, Distributed Systems, Probability & Statistics

Competitive Programming — Modern C++

- **LeetCode Knight:** Max rating **2048**, **711+** problems, 32 contests — DSA implemented primarily in **C++** (profile)
- **Codeforces Specialist:** Max rating **1419**, **252+** solved — timed contest coding in **C++** (profile)
- Focus: asymptotic complexity, memory/time trade-offs, correct fast implementations under pressure

Projects (Performance / Concurrency / Systems)

LogiFlow — Latency-Aware Large-Scale Pipeline (Google Solution Challenge 2026, Global Top 106) — *Apr 2026 – Jun 2026*

C++, Python, Redis, SQL, FastAPI, caching, simulation-style validation — GitHub | Live

- Built **high-performance** read paths (**100–400 ms**) across **580+ corridors** / **9,526 stations** by splitting hot-path **Redis** caching from heavy compute—latency vs throughput trade-off thinking.
- Modeled delay risk on **15,650** train-day records (feature design + Gradient Boosting simulator, MAE **22.7 min**); locked correctness with **100/100** automated regression checks—stress-test / simulation mindset before trusting outputs.
- Used **C++** and Python where each fit; owned modular service boundaries so performance changes stay isolatable and measurable.

AirHelp — Concurrent Real-Time Services (PowerMind Hackathon 2026) — *2026*

Python, FastAPI, WebSockets, NetworkX, A graph search, Linux deploy* — GitHub

- Shipped **concurrent** real-time services over **WebSocket** streams; profiled and tuned end-to-end latency to **1–3s** under **50–100 concurrent** connections.
- Implemented graph-based **A*** navigation (algorithm design + complexity-aware coding) with **95%+** path accuracy on custom walking graphs—DSA applied to a live networked system.

Community Hero — Production Civic Reporting PWA (Vibe2Ship 2026, Global Top 20) — *Jun 2026*

TypeScript, React, Express, Firebase, GCP Cloud Run, REST APIs — GitHub | Live

- Solo-shipped a production web service on **Linux**/Cloud Run under tight deadlines—API design, auth, geo workflows, and deploy; practiced owning an end-to-end system in production.
- Quickly learned an unfamiliar stack (Gemini, Maps) and delivered reliable user-facing behavior under ambiguous, fast-paced constraints.

Experience

Software Engineering Intern

IFFCO (Indian Farmers Fertiliser Cooperative), Phulpur Unit, Prayagraj, India — *Jun 2025 – Jul 2025*

- Designed, developed, **tested**, and deployed production services on **Linux** (Docker, CI/CD): **10+ REST APIs**, optimized SQL, structured logging, and production **debugging**.
- Improved **reliability** through unit/integration tests and peer **code review**; communicated design trade-offs clearly in a collaborative engineering environment.

Technical Skills

Modern C++ / Linux: C++17, STL, OOP, g++, debugging, Git; develop and run software on **Linux**
Role-critical CS: Data Structures & Algorithms, Operating Systems, Computer Organization & Architecture, complexity analysis
Performance & systems: Latency optimization, caching (Redis), concurrency, real-time networking (WebSockets / TCP-style streams), profiling mindset
Also: Python, C, SQL, Docker, CI/CD, pytest, probability & statistics foundations

Leadership

- **Google Solution Challenge 2026 — Global Top 106** (LogiFlow); **Vibe2Ship 2026 — Global Top 20** (Community Hero)
- **Executive Member, ACM SVNIT** & **Mentor, Nexus SVNIT** — DSA workshops and contest mentoring in **C++**